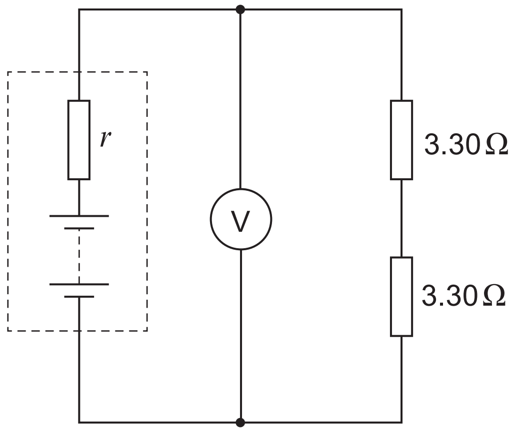
Answer **all** *questions.*

1. *(a)* Two 3.30 Ω resistors are connected in series across a battery of emf 4.80 V as shown. The voltmeter reads 4.33 V.

Diagram 1



(i) State in terms of work or energy what is meant by the emf of a battery. [2]

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(ii) Show that the internal resistance, r, of the battery is approximately 0.7 Ω. [2]

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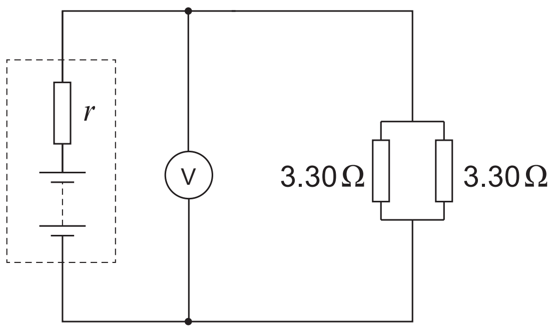
(iii) When the resistors are connected in parallel as shown below, the voltmeter reads 3.35 V.

Diagram 2



Examiner only

(I) **Without further calculation** explain why you would expect the reading to be lower, now that there is a lower resistance across the cell terminals. [2]

(II) Calculate the number of electrons entering **either** of the resistors (shown in Diagram 2) per **minute**. [3]

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(b) The heating element (a coil of wire) in an electric heater dissipates energy at a rate of 1.00 kW, when connected across the 230 V mains supply. Calculate:

(i) the resistance of the coil; [2]

(ii) the energy dissipated per hour, giving your answer in megajoules (MJ). [1]

Examiner only

(c) For each megajoule of heat from an electric heater, approximately 0.08 m^3 of gas would have to be burned in a gas-fired electricity power station. For each megajoule of heat from a domestic gas fire or boiler, approximately 0.03 m^3 of gas is burned. Discuss whether the use of electric heaters in houses should be discouraged. Calculations are not required. [3]

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